

Ejercicio 1: VLSM y enrutamiento estático en GNS3

TABLA VLSM

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Ejercicio 1: VLSM y enrutamiento estatico en GNS3

Requerimientos

LAN Administracion : 45 hosts $2^6 = 64$, $32 - 6 = 126$

LAN Ventas : 30 hosts $2^5 = 32$, $32 - 6 = 127$

LAN Soporte : 14 hosts $2^4 = 16$, $32 - 4 = 128$

LAN Gerencia : 10 hosts $2^4 = 16$, $32 - 4 = 128$

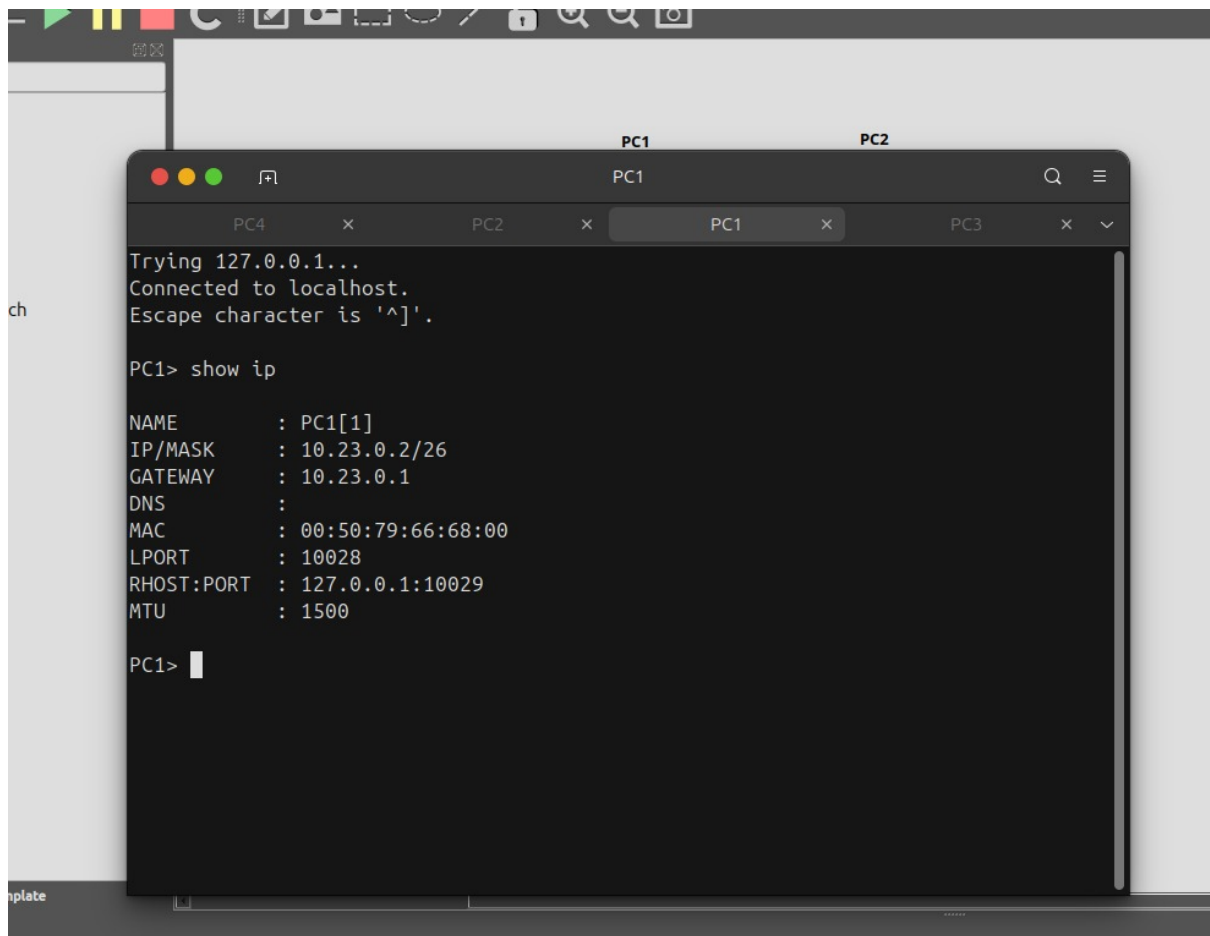
R1-R2 : 2 hosts $2^2 = 4$, $32 - 2 = 130$

R2-R3 : 2 hosts $2^2 = 4$, $32 - 2 = 130$

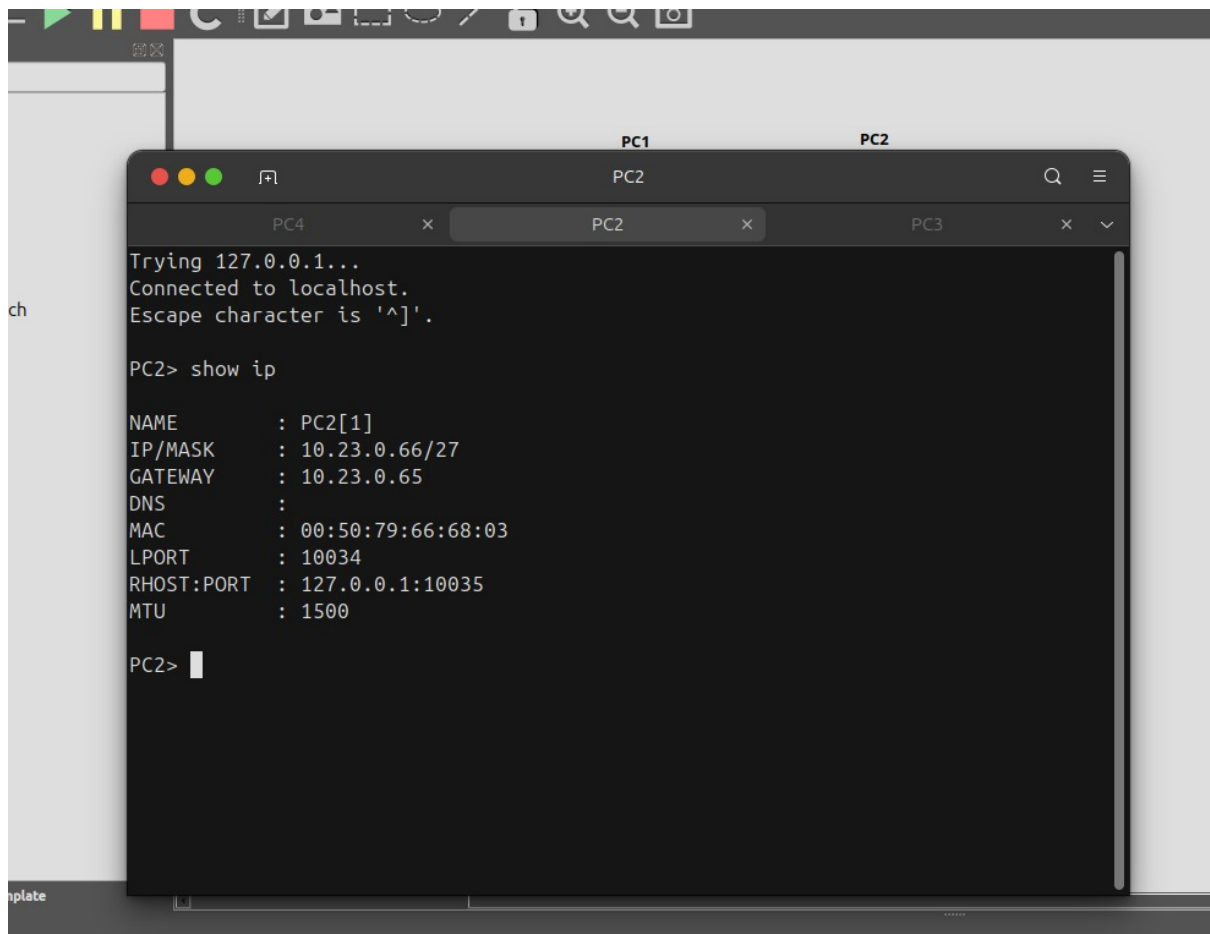
Red	Red 1	Mascara	Net	Primera IP	Ultima IP	Broadcast
LAN Adminis...	10.23.0.0	255.255.255.192	10.23.0.1	10.23.0.62	10.23.0.63	
LAN Ventas	10.23.0.64	255.255.255.128	10.23.0.65	10.23.0.94	10.23.0.95	
LAN Soporte	10.23.0.96	255.255.255.240	10.23.0.97	10.23.0.110	10.23.0.111	
LAN Gerencia	10.23.0.112	255.255.255.240	10.23.0.113	10.23.0.126	10.23.0.127	
R1-R2	10.23.0.128	255.255.255.252	10.23.0.129	10.23.0.130	10.23.0.131	
R2-R3	10.23.0.132	255.255.255.252	10.23.0.133	10.23.0.134	10.23.0.135	

Configuracion IP de cada PC:

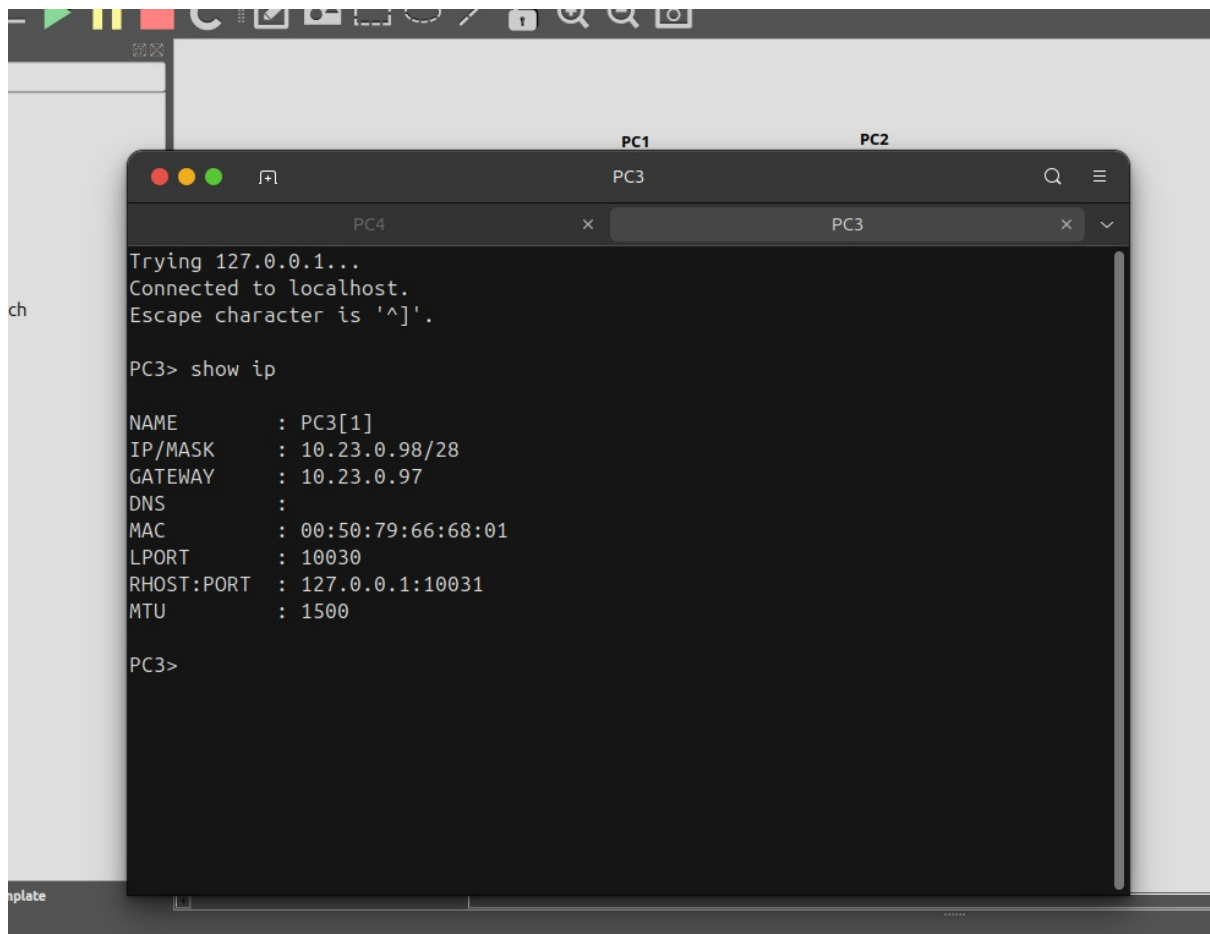
PC1



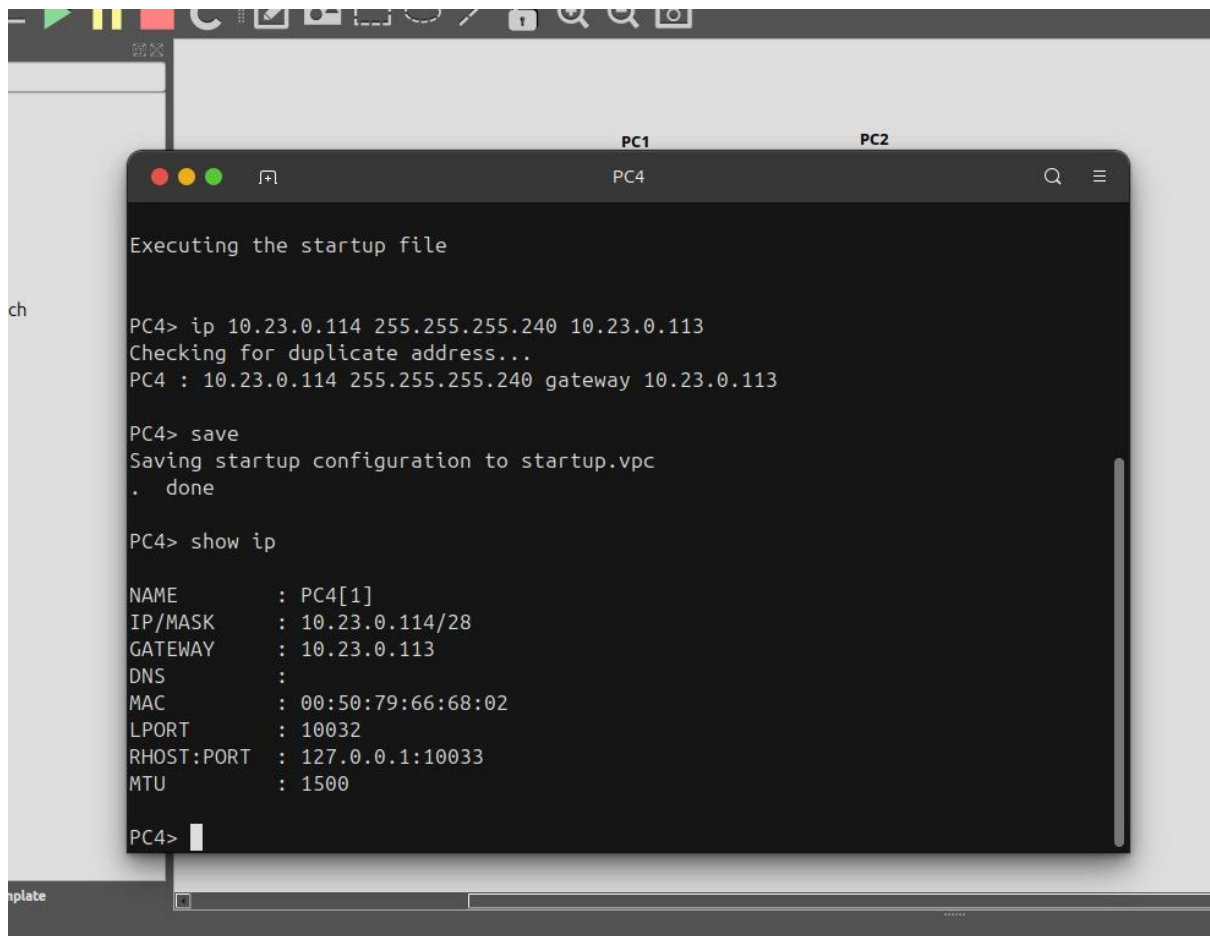
PC2



PC3:



PC4



The screenshot shows a terminal window titled 'PC4' within a network simulator environment. The window has a dark background and displays the following text:

```
Executing the startup file

PC4> ip 10.23.0.114 255.255.255.240 10.23.0.113
Checking for duplicate address...
PC4 : 10.23.0.114 255.255.255.240 gateway 10.23.0.113

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> show ip

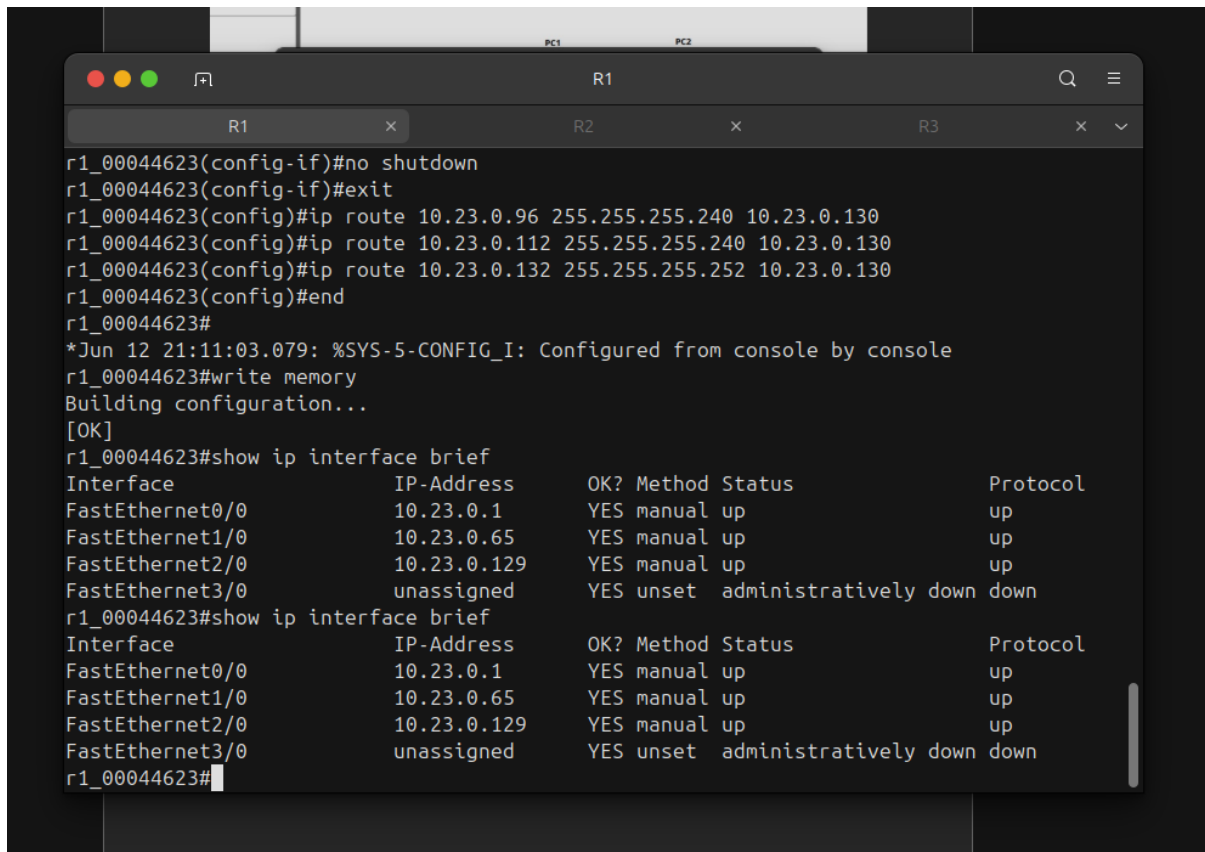
NAME       : PC4[1]
IP/MASK    : 10.23.0.114/28
GATEWAY    : 10.23.0.113
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC4> 
```

The terminal window is overlaid on a larger window showing a network topology with nodes labeled PC1, PC2, and PC4. The background window has a light gray interface with various icons and a search bar.

Configuración de interfaces de los routers

R1



The screenshot shows a terminal window for router R1. The terminal has tabs for R1, R2, and R3, with R1 selected. The commands entered are: `no shutdown`, `exit`, three `ip route` commands for destinations 10.23.0.96, 10.23.0.112, and 10.23.0.132, followed by `end`, `write memory`, and `show ip interface brief`. The output shows the configuration is saved and the interface status is displayed in a table.

```
r1_00044623(config-if)#no shutdown
r1_00044623(config-if)#exit
r1_00044623(config)#ip route 10.23.0.96 255.255.255.240 10.23.0.130
r1_00044623(config)#ip route 10.23.0.112 255.255.255.240 10.23.0.130
r1_00044623(config)#ip route 10.23.0.132 255.255.255.252 10.23.0.130
r1_00044623(config)#end
r1_00044623#
*Jun 12 21:11:03.079: %SYS-5-CONFIG_I: Configured from console by console
r1_00044623#write memory
Building configuration...
[OK]
r1_00044623#show ip interface brief
```

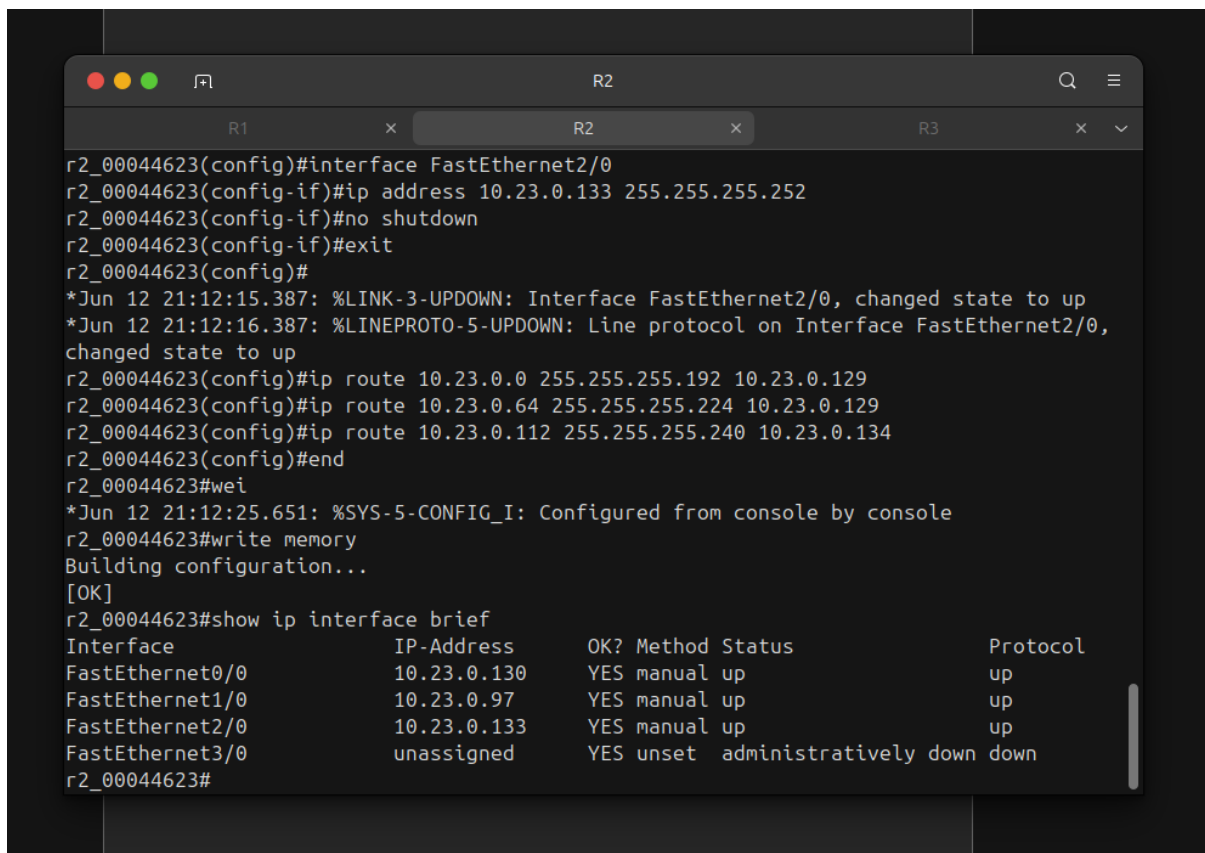
Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.23.0.1	YES	manual	up	up
FastEthernet1/0	10.23.0.65	YES	manual	up	up
FastEthernet2/0	10.23.0.129	YES	manual	up	up
FastEthernet3/0	unassigned	YES	unset	administratively down	down

```
r1_00044623#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.23.0.1	YES	manual	up	up
FastEthernet1/0	10.23.0.65	YES	manual	up	up
FastEthernet2/0	10.23.0.129	YES	manual	up	up
FastEthernet3/0	unassigned	YES	unset	administratively down	down

```
r1_00044623#
```

R2



The screenshot shows a terminal window for router R2. The terminal has tabs for R1, R2, and R3, with R2 selected. The commands entered are: `interface FastEthernet2/0`, `ip address 10.23.0.133 255.255.255.252`, `no shutdown`, `exit`, three `ip route` commands for destinations 10.23.0.0, 10.23.0.64, and 10.23.0.112, followed by `end`, `write memory`, and `show ip interface brief`. The output shows the configuration is saved and the interface status is displayed in a table.

```
r2_00044623(config)#interface FastEthernet2/0
r2_00044623(config-if)#ip address 10.23.0.133 255.255.255.252
r2_00044623(config-if)#no shutdown
r2_00044623(config-if)#exit
r2_00044623(config)#
*Jun 12 21:12:15.387: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Jun 12 21:12:16.387: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0,
changed state to up
r2_00044623(config)#ip route 10.23.0.0 255.255.255.192 10.23.0.129
r2_00044623(config)#ip route 10.23.0.64 255.255.255.224 10.23.0.129
r2_00044623(config)#ip route 10.23.0.112 255.255.255.240 10.23.0.134
r2_00044623(config)#end
r2_00044623#wei
*Jun 12 21:12:25.651: %SYS-5-CONFIG_I: Configured from console by console
r2_00044623#write memory
Building configuration...
[OK]
r2_00044623#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.23.0.130	YES	manual	up	up
FastEthernet1/0	10.23.0.97	YES	manual	up	up
FastEthernet2/0	10.23.0.133	YES	manual	up	up
FastEthernet3/0	unassigned	YES	unset	administratively down	down

```
r2_00044623#
```

R3

```
r1_00044623#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
FastEthernet0/0 10.23.0.1       YES manual up      up

r3_00044623(config-if)#ip address 10.23.0.113 255.255.255.240
r3_00044623(config-if)#no shutdown
r3_00044623(config-if)#exit
r3_00044623(config)#
*Jun 12 21:14:17.311: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Jun 12 21:14:18.311: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0,
changed state to up
r3_00044623(config)#ip route 10.23.0.0 255.255.255.192 10.23.0.133
r3_00044623(config)#ip route 10.23.0.64 255.255.255.224 10.23.0.133
r3_00044623(config)#ip route 10.23.0.96 255.255.255.240 10.23.0.133
r3_00044623(config)#ip route 10.23.0.128 255.255.255.252 10.23.0.133
r3_00044623(config)#end
r3_00044623#
*Jun 12 21:14:32.123: %SYS-5-CONFIG_I: Configured from console by console
r3_00044623#write memory
Building configuration...
[OK]
r3_00044623#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
FastEthernet0/0 10.23.0.134     YES manual up      up
FastEthernet1/0 10.23.0.113     YES manual up      up
FastEthernet2/0 unassigned      YES unset  administratively down down
FastEthernet3/0 unassigned      YES unset  administratively down down
r3_00044623#
```

Tabla de rutas estáticas configuradas.

R1

```
R1
r1_00044623#
r1_00044623#
r1_00044623#
r1_00044623#
r1_00044623#
r1_00044623#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
C       10.23.0.0/26 is directly connected, FastEthernet0/0
C       10.23.0.64/27 is directly connected, FastEthernet1/0
S       10.23.0.112/28 [1/0] via 10.23.0.130
S       10.23.0.96/28 [1/0] via 10.23.0.130
S       10.23.0.132/30 [1/0] via 10.23.0.130
C       10.23.0.128/30 is directly connected, FastEthernet2/0
r1_00044623#
```

R2

```
R2
r2_00044623#
r2_00044623#
r2_00044623#
r2_00044623#
r2_00044623#
r2_00044623#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.23.0.0/26 [1/0] via 10.23.0.129
S       10.23.0.64/27 [1/0] via 10.23.0.129
S       10.23.0.112/28 [1/0] via 10.23.0.134
S       10.23.0.96/28 is directly connected, FastEthernet1/0
C       10.23.0.132/30 is directly connected, FastEthernet2/0
C       10.23.0.128/30 is directly connected, FastEthernet0/0
r2_00044623#
```

R3

```
r3_00044623(config-if)#ip address 10.23.0.113 255.255.255.240
r3_00044623(config-if)#no shutdown
r3_00044623(config-if)#exit
r3_00044623(config)#
*Jun 12 21:14:17.311: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Jun 12 21:14:18.311: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0,
changed state to up
r3_00044623(config)#ip route 10.23.0.0 255.255.255.192 10.23.0.133
r3_00044623(config)#ip route 10.23.0.64 255.255.255.224 10.23.0.133
r3_00044623(config)#ip route 10.23.0.96 255.255.255.240 10.23.0.133
r3_00044623(config)#ip route 10.23.0.128 255.255.255.252 10.23.0.133
r3_00044623(config)#end
r3_00044623#
*Jun 12 21:14:32.123: %SYS-5-CONFIG_I: Configured from console by console
r3_00044623#write memory
Building configuration...
[OK]
r3_00044623#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.23.0.134	YES	manual	up	up
FastEthernet1/0	10.23.0.113	YES	manual	up	up
FastEthernet2/0	unassigned	YES	unset	administratively down	down
FastEthernet3/0	unassigned	YES	unset	administratively down	down

```
r3_00044623#
2/28 [i/o] via 10.23.0.134
2/28 is directly connected, FastEthernet1/0
2/30 is directly connected, FastEthernet2/0
8/30 is directly connected, FastEthernet0/0
```

PRUEBA DE LA PRIEMRA A LA MAS LEJANA

```
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
PC1> ping 10.23.0.114
84 bytes from 10.23.0.114 icmp_seq=1 ttl=61 time=69.574 ms
84 bytes from 10.23.0.114 icmp_seq=2 ttl=61 time=36.363 ms
84 bytes from 10.23.0.114 icmp_seq=3 ttl=61 time=35.966 ms
^C
PC1> trace 10.23.0.114
trace to 10.23.0.114, 8 hops max, press Ctrl+C to stop
1 10.23.0.1 9.407 ms 9.882 ms 9.411 ms
2 10.23.0.130 30.318 ms 19.980 ms 20.094 ms
3 10.23.0.134 40.314 ms 29.529 ms 30.332 ms
4 *10.23.0.114 50.243 ms (ICMP type:3, code:3, Destination port unreachable)
PC1>
```

DE PC1 A PC4

Ejercicio 2: VLSM y enrutamiento dinámico OSPF en GNS3

Tabla VLSM

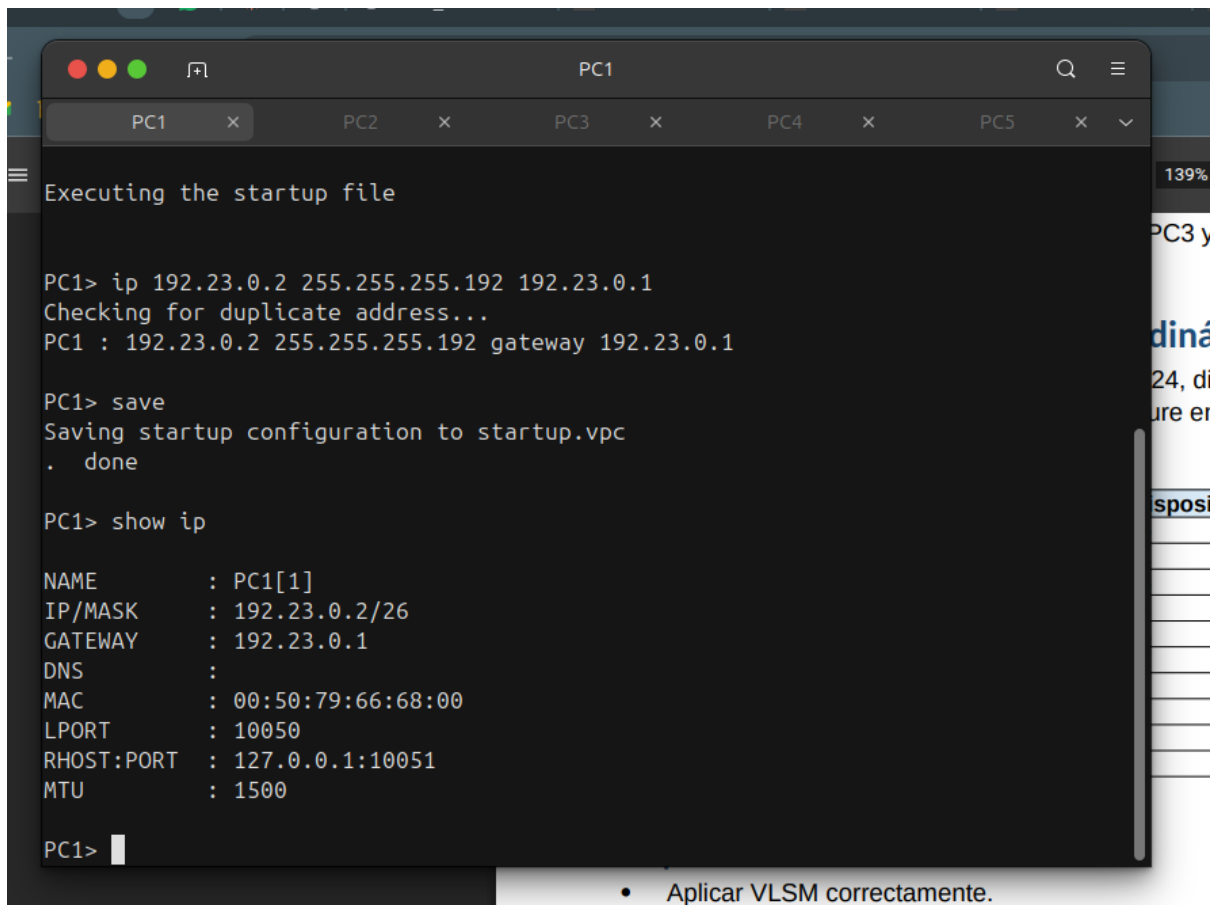
LAN PC1	40 hosts	$2^6 = 64$	126
LAN PC2	28	$2^5 = 32$	127
LAN PC3	18	$2^5 = 32$	127
LAN PC4	9	$2^4 = 16$	128
LAN PC5	5	$2^3 = 8$	129
R1-R2	2	$2^2 = 4$	130
R1-R3	2	$2^2 = 4$	130

R2-R5	2	$2^2 = 4$	130
R3-R4	2	$2^2 = 4$	130
R4-R5	2	$2^2 = 4$	130

RED	Red	Mascara	Primera IP	Ultima IP	Broadcast
LAN PC1 R5	192.23.0.0	255.255.255.192	192.23.0.1	192.23.0.62	192.23.0.63
LAN PC2 R2	192.23.0.64	255.255.255.224	192.23.0.65	192.23.0.94	192.23.0.95
LAN PC3 R3	192.23.0.96	255.255.255.224	192.23.0.97	192.23.0.126	192.23.0.127
LAN PC4 R1	192.23.0.128	255.255.255.240	192.23.0.129	192.23.0.142	192.23.0.143
LAN PC5 R4	192.23.0.144	255.255.255.248	192.23.0.145	192.23.0.150	192.23.0.151
R1-R2	192.23.0.152	255.255.255.252	192.23.0.153	192.23.0.154	192.23.0.155
R1-R3	192.23.0.156	255.255.255.252	192.23.0.157	192.23.0.158	192.23.0.159
R2-R5	192.23.0.160	255.255.255.252	192.23.0.161	192.23.0.162	192.23.0.163
R3-R4	192.23.0.164	255.255.255.252	192.23.0.165	192.23.0.166	192.23.0.167
R4-R5	192.23.0.168	255.255.255.252	192.23.0.169	192.23.0.170	192.23.0.171

Tabla de asignación IP por dispositivo.

PC1



The screenshot shows a terminal window titled 'PC1' with tabs for PC1, PC2, PC3, PC4, and PC5. The terminal output is as follows:

```
Executing the startup file

PC1> ip 192.23.0.2 255.255.255.192 192.23.0.1
Checking for duplicate address...
PC1 : 192.23.0.2 255.255.255.192 gateway 192.23.0.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

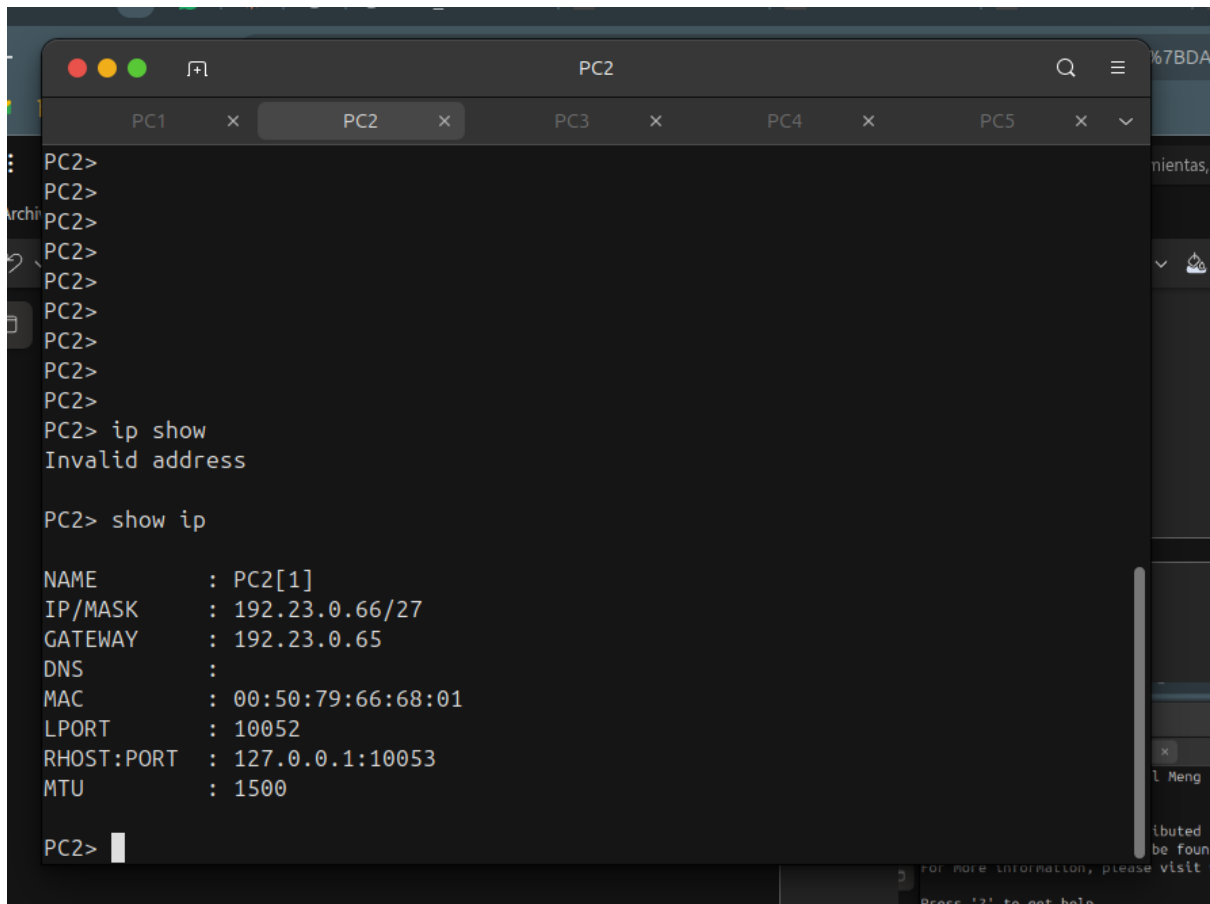
NAME       : PC1[1]
IP/MASK    : 192.23.0.2/26
GATEWAY    : 192.23.0.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10050
RHOST:PORT : 127.0.0.1:10051
MTU        : 1500

PC1> 
```

Below the terminal window, there is a bullet point:

- Aplicar VLSM correctamente.

PC2

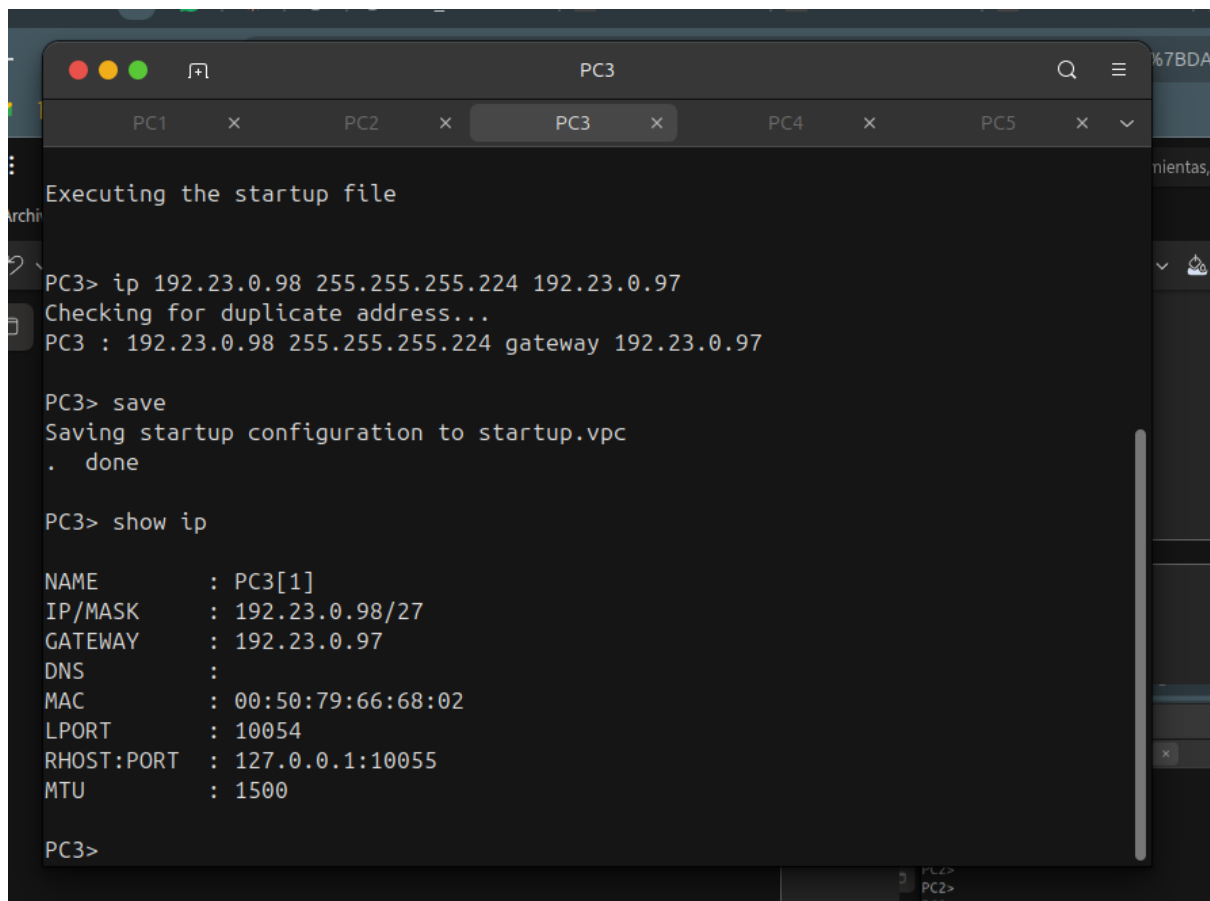


The image shows a terminal window titled "PC2" with a tab bar at the top containing tabs for PC1, PC2, PC3, PC4, and PC5. The terminal displays a series of commands and their outputs:

```
PC2>  
PC2>  
PC2>  
PC2>  
PC2>  
PC2>  
PC2>  
PC2>  
PC2>  
PC2> ip show  
Invalid address  
  
PC2> show ip  
  
NAME       : PC2[1]  
IP/MASK     : 192.23.0.66/27  
GATEWAY     : 192.23.0.65  
DNS         :  
MAC        : 00:50:79:66:68:01  
LPORT      : 10052  
RHOST:PORT  : 127.0.0.1:10053  
MTU        : 1500  
  
PC2> |
```

At the bottom of the terminal, there is a footer that reads: "For more information, please visit [https://www.powershell.com](#) Press '?' to get help."

PC3



PC4

The image shows a terminal window titled 'PC4' with tabs for PC1, PC2, PC3, PC4, and PC5. The terminal displays the following commands and output:

```
Executing the startup file
PC4> ip 192.23.0.130 255.255.255.240 192.23.0.129
Checking for duplicate address...
PC4 : 192.23.0.130 255.255.255.240 gateway 192.23.0.129

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> show ip

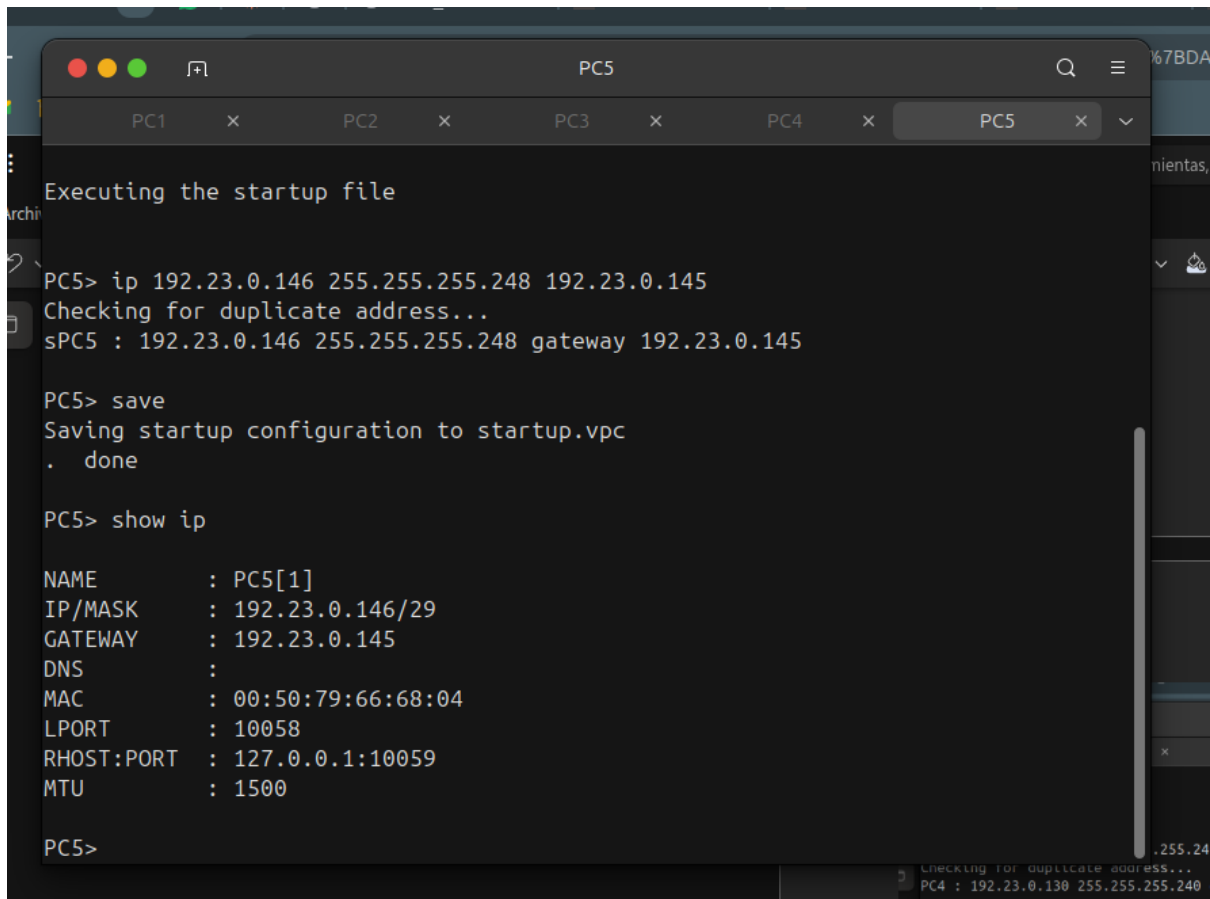
NAME       : PC4[1]
IP/MASK    : 192.23.0.130/28
GATEWAY    : 192.23.0.129
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10056
RHOST:PORT : 127.0.0.1:10057
MTU        : 1500

PC4>
```

At the bottom of the window, a status bar shows the configuration for PC3[1]:

NAME	: PC3[1]
IP/MASK	: 192.23.0.98/27

PC5



```
PC5
PC1 x PC2 x PC3 x PC4 x PC5 x
Executing the startup file
PC5> ip 192.23.0.146 255.255.255.248 192.23.0.145
Checking for duplicate address...
sPC5 : 192.23.0.146 255.255.255.248 gateway 192.23.0.145

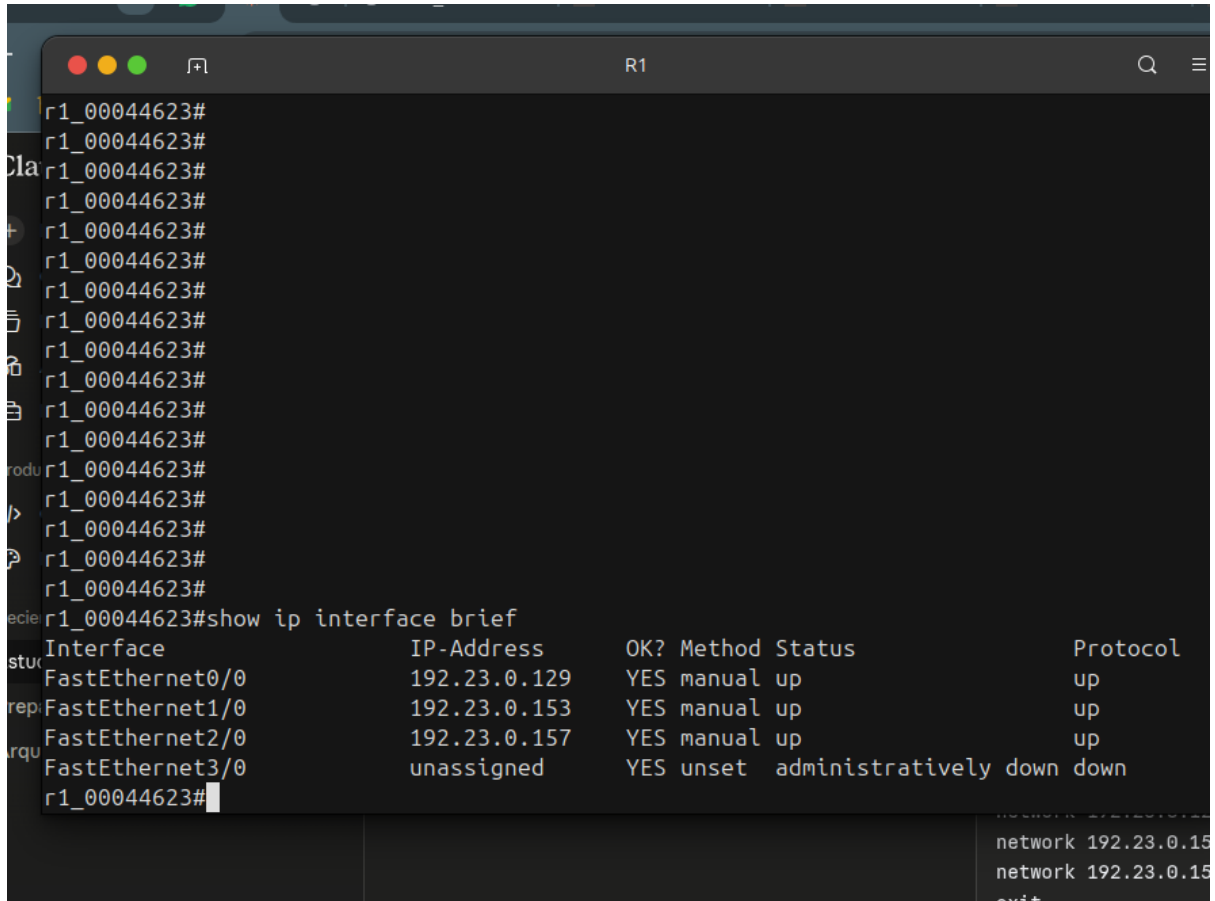
PC5> save
Saving startup configuration to startup.vpc
. done

PC5> show ip

NAME       : PC5[1]
IP/MASK     : 192.23.0.146/29
GATEWAY     : 192.23.0.145
DNS         :
MAC         : 00:50:79:66:68:04
LPORT      : 10058
RHOST:PORT  : 127.0.0.1:10059
MTU         : 1500

PC5>
```

R1



R2

```
R2
R1 x R2 x
*Jun 12 22:20:55.079: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Jun 12 22:20:56.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0,
changed state to up
r2_00044623(config)#router ospf 1
r2_00044623(config-router)#network 192.23.0.152 0.0.0.3 area 0
r2_00044623(config-router)#network 192.23.0.64 0.0.0.31 area 0
r2_00044623(config-router)#
*Jun 12 22:21:02.379: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.157 on FastEthernet0/0 f
rom LOADING to FULL, Loading Done
r2_00044623(config-router)#network 192.23.0.160 0.0.0.3 area 0
r2_00044623(config-router)#exit
r2_00044623(config)#end
r2_00044623#
*Jun 12 22:21:15.819: %SYS-5-CONFIG_I: Configured from console by console
r2_00044623#write memory
Building configuration...
[OK]
r2_00044623#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.23.0.154 YES manual up up
FastEthernet1/0 192.23.0.65 YES manual up up
FastEthernet2/0 192.23.0.161 YES manual up up
FastEthernet3/0 unassigned YES unset administratively down down
r2_00044623#
```

R3

```
R3
R1 x R2 x R3 x
r3_00044623#
*Jun 12 22:21:58.643: %SYS-5-CONFIG_I: Configured from console by console
r3_00044623#
*Jun 12 22:22:00.307: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Jun 12 22:22:00.471: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Jun 12 22:22:00.527: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Jun 12 22:22:01.307: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
r3_00044623#
*Jun 12 22:22:01.471: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0,
changed state to up
*Jun 12 22:22:01.527: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0,
changed state to up
r3_00044623#
*Jun 12 22:22:02.799: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.157 on FastEthernet0/0 f
rom LOADING to FULL, Loading Done
r3_00044623#show ip interf
r3_00044623#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.23.0.158 YES manual up up
FastEthernet1/0 192.23.0.97 YES manual up up
FastEthernet2/0 192.23.0.165 YES manual up up
FastEthernet3/0 unassigned YES unset administratively down down
r3_00044623#
router ospf 1
network 192.23.0.156
```

R4

```

R4
R1 x R2 x R3 x R4 x
r4_00044623#write memory
*Jun 12 22:22:38.303: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Jun 12 22:22:38.459: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Jun 12 22:22:38.523: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Jun 12 22:22:39.303: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
r4_00044623#write memory
*Jun 12 22:22:39.459: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0,
changed state to up
*Jun 12 22:22:39.527: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0,
changed state to up
r4_00044623#write memory
Building configuration...
[OK]
r4_00044623#
*Jun 12 22:22:45.791: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.165 on FastEthernet0/0 f
rom LOADING to FULL, Loading Done
r4_00044623#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.23.0.166 YES manual up up
FastEthernet1/0 192.23.0.145 YES manual up up
FastEthernet2/0 192.23.0.169 YES manual up up
FastEthernet3/0 unassigned YES unset administratively down down
r4_00044623#
```

R5

```
rom LOADING to FULL, Loading Done
r5_00044623#write memory
*Jun 12 22:24:11.851: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Jun 12 22:24:12.019: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Jun 12 22:24:12.039: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Jun 12 22:24:12.851: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
r5_00044623#write memory
Building configuration...

*Jun 12 22:24:13.019: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0,
changed state to up
*Jun 12 22:24:13.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0,
changed state to up[OK]
r5_00044623#
*Jun 12 22:24:16.499: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.161 on FastEthernet0/0 f
rom LOADING to FULL, Loading Done
r5_00044623#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          192.23.0.162    YES manual up              up
FastEthernet1/0          192.23.0.170    YES manual up              up
FastEthernet2/0          192.23.0.1      YES manual up              up
FastEthernet3/0          unassigned      YES unset  administratively down down
r5_00044623#
```

Configuración OSPF.

R1


```
R3
R3 x R4 x R5 x
*Jun 13 23:12:09.175: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Jun 13 23:12:09.175: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Jun 13 23:12:09.543: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0
, changed state to up
*Jun 13 23:12:09.547: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0
, changed state to up
*Jun 13 23:12:09.551: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0
, changed state to up
*Jun 13 23:12:09.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet3/0
, changed state to down
*Jun 13 23:12:10.739: %LINK-5-CHANGED: Interface FastEthernet3/0, changed state to ad
ministratively down
*Jun 13 23:12:49.575: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.169 on FastEthernet2/0
from LOADING to FULL, Loading Done
*Jun 13 23:12:54.331: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.157 on FastEthernet0/0
from LOADING to FULL, Loading Done
r3_00044623#
r3_00044623#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 192.23.0.96 0.0.0.31 area 0
 network 192.23.0.156 0.0.0.3 area 0
 network 192.23.0.164 0.0.0.3 area 0
r3_00044623#
```

R4

```
R4
R3 x R4 x R5 x
a cold start
*Jun 13 23:12:09.367: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Jun 13 23:12:09.367: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Jun 13 23:12:09.707: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0
, changed state to up
*Jun 13 23:12:09.711: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0
, changed state to up
*Jun 13 23:12:09.715: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0
, changed state to up
*Jun 13 23:12:09.723: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet3/0
, changed state to down
*Jun 13 23:12:10.903: %LINK-5-CHANGED: Interface FastEthernet3/0, changed state to ad
ministratively down
*Jun 13 23:12:49.419: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.165 on FastEthernet0/0
from LOADING to FULL, Loading Done
*Jun 13 23:12:54.203: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.0.170 on FastEthernet2/0
from LOADING to FULL, Loading Done
r4_00044623#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 192.23.0.144 0.0.0.7 area 0
 network 192.23.0.164 0.0.0.3 area 0
 network 192.23.0.168 0.0.0.3 area 0
r4_00044623#
```

R5

R2

```
R2
r2_00044623#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.0.0/24 is variably subnetted, 10 subnets, 5 masks
O       192.23.0.144/29 [110/3] via 192.23.0.162, 00:17:07, FastEthernet2/0
O       192.23.0.156/30 [110/2] via 192.23.0.153, 00:17:07, FastEthernet0/0
C       192.23.0.152/30 is directly connected, FastEthernet0/0
O       192.23.0.128/28 [110/2] via 192.23.0.153, 00:17:07, FastEthernet0/0
O       192.23.0.164/30 [110/3] via 192.23.0.162, 00:17:07, FastEthernet2/0
        [110/3] via 192.23.0.153, 00:17:07, FastEthernet0/0
C       192.23.0.160/30 is directly connected, FastEthernet2/0
O       192.23.0.168/30 [110/2] via 192.23.0.162, 00:17:17, FastEthernet2/0
C       192.23.0.64/27 is directly connected, FastEthernet1/0
O       192.23.0.96/27 [110/3] via 192.23.0.153, 00:17:07, FastEthernet0/0
O       192.23.0.0/26 [110/2] via 192.23.0.162, 00:17:17, FastEthernet2/0
r2_00044623#
```

R3

```
R3
r3_00044623#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.0.0/24 is variably subnetted, 10 subnets, 5 masks
O       192.23.0.144/29 [110/2] via 192.23.0.166, 00:17:23, FastEthernet2/0
C       192.23.0.156/30 is directly connected, FastEthernet0/0
O       192.23.0.152/30 [110/2] via 192.23.0.157, 00:17:13, FastEthernet0/0
O       192.23.0.128/28 [110/2] via 192.23.0.157, 00:17:13, FastEthernet0/0
C       192.23.0.164/30 is directly connected, FastEthernet2/0
O       192.23.0.160/30 [110/3] via 192.23.0.166, 00:17:13, FastEthernet2/0
        [110/3] via 192.23.0.157, 00:17:13, FastEthernet0/0
O       192.23.0.168/30 [110/2] via 192.23.0.166, 00:17:13, FastEthernet2/0
O       192.23.0.64/27 [110/3] via 192.23.0.157, 00:17:13, FastEthernet0/0
C       192.23.0.96/27 is directly connected, FastEthernet1/0
O       192.23.0.0/26 [110/3] via 192.23.0.166, 00:17:13, FastEthernet2/0
r3_00044623#
```

R4

```
r4_00044623#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.0.0/24 is variably subnetted, 10 subnets, 5 masks
C       192.23.0.144/29 is directly connected, FastEthernet1/0
O       192.23.0.156/30 [110/2] via 192.23.0.165, 00:17:27, FastEthernet0/0
O       192.23.0.152/30 [110/3] via 192.23.0.170, 00:17:17, FastEthernet2/0
        [110/3] via 192.23.0.165, 00:17:17, FastEthernet0/0
O       192.23.0.128/28 [110/3] via 192.23.0.165, 00:17:17, FastEthernet0/0
C       192.23.0.164/30 is directly connected, FastEthernet0/0
O       192.23.0.160/30 [110/2] via 192.23.0.170, 00:17:17, FastEthernet2/0
C       192.23.0.168/30 is directly connected, FastEthernet2/0
O       192.23.0.64/27 [110/3] via 192.23.0.170, 00:17:17, FastEthernet2/0
O       192.23.0.96/27 [110/2] via 192.23.0.165, 00:17:27, FastEthernet0/0
O       192.23.0.0/26 [110/2] via 192.23.0.170, 00:17:17, FastEthernet2/0
r4_00044623#
```

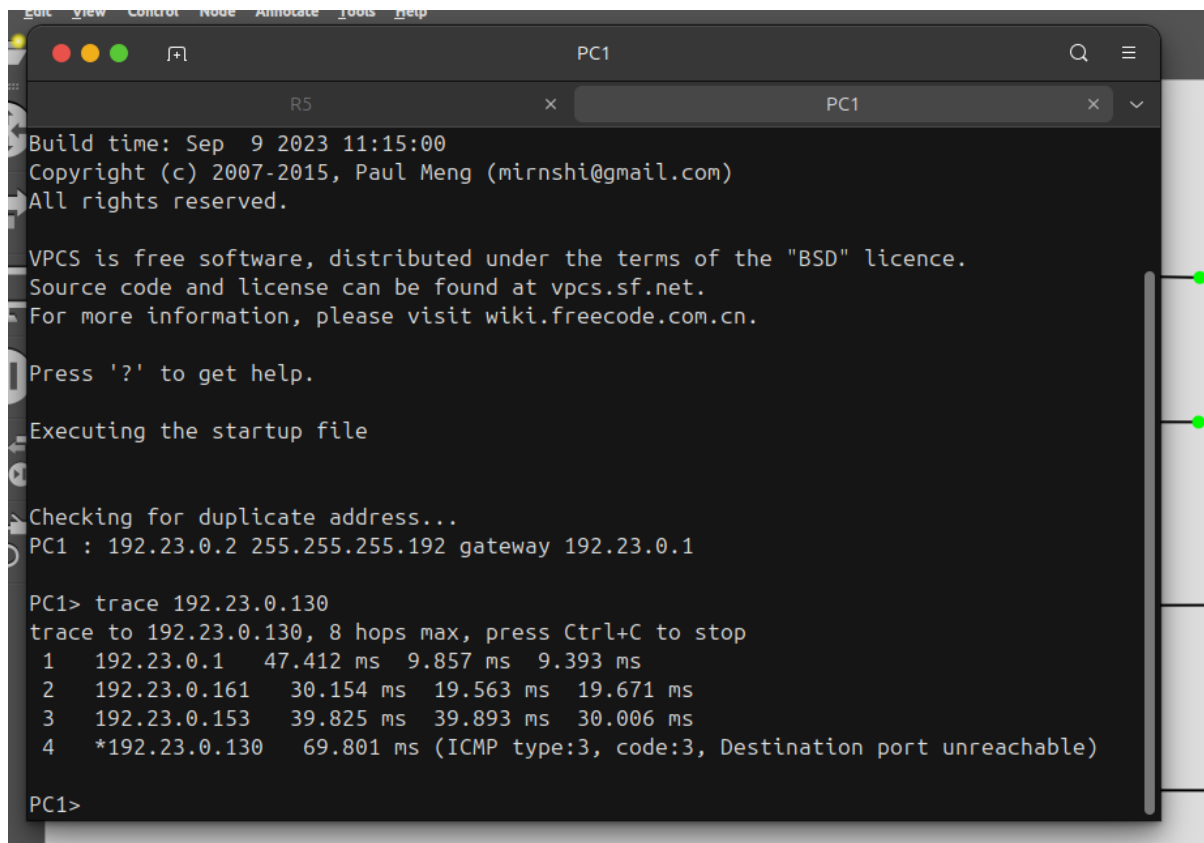
R5

```
r5_00044623#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.0.0/24 is variably subnetted, 10 subnets, 5 masks
O       192.23.0.144/29 [110/2] via 192.23.0.169, 00:17:22, FastEthernet1/0
O       192.23.0.156/30 [110/3] via 192.23.0.169, 00:17:22, FastEthernet1/0
        [110/3] via 192.23.0.161, 00:17:22, FastEthernet0/0
O       192.23.0.152/30 [110/2] via 192.23.0.161, 00:17:32, FastEthernet0/0
O       192.23.0.128/28 [110/3] via 192.23.0.161, 00:17:22, FastEthernet0/0
O       192.23.0.164/30 [110/2] via 192.23.0.169, 00:17:22, FastEthernet1/0
C       192.23.0.160/30 is directly connected, FastEthernet0/0
C       192.23.0.168/30 is directly connected, FastEthernet1/0
O       192.23.0.64/27 [110/2] via 192.23.0.161, 00:17:32, FastEthernet0/0
O       192.23.0.96/27 [110/3] via 192.23.0.169, 00:17:22, FastEthernet1/0
C       192.23.0.0/26 is directly connected, FastEthernet2/0
r5_00044623#
```

PRUEBA DE LA PRIMERA A LA MAS LEJANA (PC1 – PC5)



The screenshot shows a VPCS terminal window titled "PC1". The window has a menu bar with "Edit", "View", "Control", "Node", "Annotate", "Tools", and "Help". Below the menu bar, there are tabs for "R5" and "PC1". The terminal output is as follows:

```
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
PC1 : 192.23.0.2 255.255.255.192 gateway 192.23.0.1

PC1> trace 192.23.0.130
trace to 192.23.0.130, 8 hops max, press Ctrl+C to stop
 1  192.23.0.1    47.412 ms  9.857 ms  9.393 ms
 2  192.23.0.161  30.154 ms  19.563 ms  19.671 ms
 3  192.23.0.153  39.825 ms  39.893 ms  30.006 ms
 4  *192.23.0.130 69.801 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>
```